

# Vaibhav KRISHAN

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## RESEARCH POSITIONS

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| <b>PDF (TCS)</b> | The Institute of Mathematical Sciences | July 2024-Present |
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## EDUCATION

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| <b>Ph.D.</b><br>COMPUTER SCIENCE<br>AND ENGINEERING | Indian Institute of Technology Bombay<br>Advised by Prof. Sundar Vishwanathan<br>and Prof. Nutan Limaye | July 2017-May 2024 |
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| <b>B.Tech.</b><br>COMPUTER SCIENCE<br>AND ENGINEERING | Indian Institute of Technology Bombay<br>CPI: 7.59/10 | July 2009-May 2013 |
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## JOURNAL PUBLICATIONS

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| <b>Algorithmica</b> | A #SAT Algorithm for Small Constant-depth Circuits with PTF gates with S. Bajpai, D. Kush, N. Limaye and S. Srinivasan<br>Algorithmica 84, 1132-1162 (2022). |
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## CONFERENCE PUBLICATIONS

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| <b>MFCS 2026</b> | On $CC^0$ Lower Bounds for AND via Torus Polynomials with J. Sarma<br>The 51st International Symposium on Mathematical Foundations of Computer Science, 2026 (to appear).                                    |
| <b>SAT 2026</b>  | On Proof Systems for #QBF with S. Chede, L. Chew and A. Shukla<br>The 29th International Conference on Theory and Applications of Satisfiability Testing, 2026 (to appear).                                  |
| <b>ITCS 2026</b> | Lower Bounds and Separations for Torus Polynomials with S. Vishwanathan<br>The 17 <sup>th</sup> Innovations in Theoretical Computer Science Conference, 2026.                                                |
| <b>CSR 2021</b>  | Upper Bound for Torus Polynomials<br>The 16th International Computer Science Symposium in Russia, 2021.                                                                                                      |
| <b>ITCS 2019</b> | A #SAT Algorithm for Small Constant-depth Circuits with PTF gates with S. Bajpai, D. Kush, N. Limaye and S. Srinivasan<br>The 10 <sup>th</sup> Innovations in Theoretical Computer Science Conference, 2019. |

## PREPRINTS

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**ECCC**

Isolation Lemma for Directed Reachability and NL vs. L  
with N. Limaye

## TALKS AND PRESENTATIONS

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**Presentation**

Lower Bounds and Separations for Torus Polynomials  
ITCS 2026.

**Video**

Lower Bounds and Separations for Torus Polynomials  
ITCS 2026.

**Lecture**

Polynomials and Computation  
Foundational Lecture Series on TCS, IMSc, 2025.

**Presentation**

Upper Bound for Torus Polynomials  
CSR 2021. (online)

**Presentation**  
and **Poster**

A #SAT Algorithm for Small Constant-depth Circuits with PTF gates  
ITCS 2019.

## SERVICE AND ENGAGEMENT

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Sub-reviewer

SAT 2026.

Sub-reviewer

Computational Complexity Conference (CCC) 2024.

Sub-reviewer

Foundations of Software Technology and Theoretical Computer Science  
(FSTTCS) 2024

Volunteer

FSTTCS 2019

## TEACHING ASSISTANCE

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- Awarded best TA of the semester for CS 101.
- Awarded best TA of the month for Automata Theory.
- TA duty for complexity theory twice, design and analysis of algorithms, combinatorics, and theoretical machine learning.

## PROFESSIONAL EXPERIENCE

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- Freelance work in quantitative trading for multiple clients since May '22.
- Quantitative Research Consultant for AlphaGrep, and for 5-Swans, during official break from PhD June '21-May '22.
- As Quantitative Developer in AlphaGrep from June '16-July 17 and Jan '14-April '15.
- As Data Scientist in Housing.com from May '15-May '16.
- As Software Engineer in Samsung from July '13-Dec '13.

## PERSONAL DETAILS

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D.O.B. 09<sup>th</sup> December 1993

Sex Male

Nationality Indian